

JONGHA (JON) RYU

Room W324 Westgate Building University Park, PA 16802

✉ jongha@psu.edu 🏠 jongharyu.github.io

RESEARCH INTERESTS	My research builds spectral and information-theoretic foundations for machine learning. As an information theorist by training, I distill complex ideas, unify principles across domains, and translate them into scalable algorithms bridging theory and practice. Recent work includes neural network-based operator learning for scientific simulation [C9], [C15], [C16], score-based generative modeling [C12], and uncertainty quantification and anytime-valid inference [J2], [C10], [C11], [C14].	
EMPLOYMENT	Pennsylvania State University Assistant Professor at Department of Computer Science and Engineering · A faculty co-hire in the Institute for Computational and Data Sciences (ICDS) · Wormley Family Early Career Professorship	Aug. 2026 – present
	Massachusetts Institute of Technology (MIT) Postdoctoral Associate at Electrical Engineering & Computer Science Department · Advisor: Prof. Gregory W. Wornell	Aug. 2022 – Aug. 2026
EDUCATION	University of California San Diego (UCSD) Ph.D. in Electrical Engineering (GPA: 3.99/4.0) · Thesis advisors: Prof. Young-Han Kim and Prof. Sanjoy Dasgupta · Thesis title: “From Information Theory to Machine Learning Algorithms: A Few Vignettes” M.S. in Electrical Engineering	Sep. 2015 – Jun. 2022 Dec. 2018
	Seoul National University (SNU) Bachelor of Science (<i>summa cum laude</i> , GPA: 4.11/4.3) · Double major in Electrical and Computer Engineering & Mathematical Sciences; minor in Physics	Mar. 2008 – Aug. 2015
JOURNAL PAPERS	(* indicates equal contribution. † indicates alphabetical orders.) [J1] J. Jon Ryu [*] , Shouvik Ganguly [*] , Young-Han Kim, Yung-Kyun Noh, Daniel Lee, “Nearest neighbor density functional estimation from inverse Laplace transform”. <i>IEEE Trans. Info. Theory</i> , vol. 68, no. 6, pp. 3511-3551, Jun. 2022. [J2] J. Jon Ryu , Alankrita Bhatt, “On Confidence Sequences for Bounded Random Processes via Universal Gambling Strategies,” <i>IEEE Trans. Info. Theory</i> , vol. 70, no. 10, pp. 7143-7161, Oct. 2024.	
CONFERENCE PAPERS	[C1] Alankrita Bhatt [†] , Jiun-Ting Huang [†] , Young-Han Kim [†] , J. Jon Ryu [†] , Pinar Sen [†] , “Monte Carlo methods for randomized likelihood decoding,” <i>Annual Allerton Conference on Communication, Control, and Computation (Allerton)</i> , Sep. 2018. [C2] Jongha Ryu , Young-Han Kim, “Conditional distribution learning using neural networks and its application to universal image denoising,” <i>IEEE International Conference on Image Processing (ICIP)</i> , Oct. 2018. [C3] Alankrita Bhatt [†] , Jiun-Ting Huang [†] , Young-Han Kim [†] , J. Jon Ryu [†] , Pinar Sen [†] , “Variations on a theme by Liu, Cuff, and Verdú: The power of posterior sampling,” <i>IEEE Information Theory Workshop (ITW)</i> , Nov. 2018. [C4] Yang Yang, Guillaume Sautiere, J. Jon Ryu , Taco Cohen, “Feedback Recurrent Autoencoder,” <i>IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)</i> , May 2020. [C5] J. Jon Ryu , Jiun-Ting Huang, Young-Han Kim, “On the Role of Eigendecomposition in Kernel Embedding,” <i>IEEE International Symposium on Information Theory (ISIT)</i> , Jun. 2021. [C6] J. Jon Ryu , Alankrita Bhatt, Young-Han Kim, “Parameter-free Online Linear Optimization with Side Information via Universal Coin Betting,” <i>International Conference on Artificial Intelligence and Statistics (AISTATS)</i> , Apr. 2022.	

- [C7] Alankrita Bhatt*, **J. Jon Ryu***, Young-Han Kim, “On Universal Portfolios with Continuous Side Information,” *AISTATS*, Apr. 2023.
- [C8] Abhin Shah, Maohao Shen, **J. Jon Ryu**, Subhro Das, Prasanna Sattigeri, Yuheng Bu, Gregory W. Wornell, “Group Fairness with Uncertainty in Sensitive Attributes,” *ISIT*, Jul. 2024. (Note: A preliminary version of this manuscript was presented at *ICML 2023 Workshop on Spurious Correlations, Invariance, and Stability*.)
- [C9] **J. Jon Ryu**, Xiangxiang Xu, Hasan Sabri Melihcan Erol, Yuheng Bu, Lizhong Zheng, Gregory Wornell, “Operator SVD with Neural Networks via Nested Low-Rank Approximation,” *International Conference on Machine Learning (ICML)*, Jul. 2024.
- [C10] **J. Jon Ryu**, Gregory W. Wornell, “Gambling-Based Confidence Sequences for Bounded Random Vectors,” *ICML*, Jul. 2024. **Spotlight** (top 3.5%).
- [C11] Maohao Shen*, **J. Jon Ryu***, Soumya Ghosh, Yuheng Bu, Prasanna Sattigeri, Subhro Das, Gregory W. Wornell, “Are Uncertainty Quantification Capabilities of Evidential Deep Learning a Mirage?,” *Annual Conference on Neural Information Processing Systems (NeurIPS)*, Dec. 2024.
- [C12] Tejas Jayashankar*, **J. Jon Ryu***, Gregory W. Wornell, “Score-of-Mixture Training: Training One-Step Generative Models Made Simple via Score Estimation of Mixture Distributions,” *ICML*, Jul. 2025. **Spotlight** (top 2.6%).
- [C13] **J. Jon Ryu**, Abhin Shah, Gregory W. Wornell, “A Unified View on Learning Unnormalized Distributions via Noise-Contrastive Estimation,” *ICML*, Jul. 2025.
- [C14] **J. Jon Ryu**, Jeongyeol Kwon, Benjamin Koppe, Kwang-Sung Jun, “Improved Offline Contextual Bandits with Second-Order Bounds: Betting and Freezing,” *Conference on Learning Theory (COLT)*, Jun. 2025.
- [C15] Minchan Jeong*, **J. Jon Ryu***, Se-Young Yun, Gregory W. Wornell, “Efficient Parametric SVD of Koopman Operator for Stochastic Dynamical Systems,” *NeurIPS*, Dec. 2025.
- [C16] **J. Jon Ryu**, Samuel Zhou, Gregory W. Wornell, “Revisiting Orbital Minimization for Neural Operator Decomposition,” *NeurIPS*, Dec. 2025.
- [C17] **J. Jon Ryu**, Pavan Yeddanapudi, Xiangxiang Xu, Gregory W. Wornell, “Contrastive Predictive Coding Done Right for Mutual Information Estimation,” *International Conference on Learning Representations (ICLR)*, Apr. 2026.
- [C18] Hasan Sabri Melihcan Erol, Suat Evren, Oktay Ozel, Alexander Morgan, **J. Jon Ryu**, Lizhong Zheng, “When Softmax Fails at the Top: Extreme-Value Corrections for InfoNCE,” *ICML*, Jul. 2026.
- [C19] Hasan Sabri Melihcan Erol, **J. Jon Ryu**, Lizhong Zheng, “Beyond Linsker’s Infomax Principle: A Rate-Distortion Perspective on InfoNCE,” *ITW*, Nov. 2026.
- [C20] Minchan Jeong*, **J. Jon Ryu***, Se-Young Yun, Gregory W. Wornell, “Variational Monte Carlo for Quantum Excited States via Nested Low-Rank Approximation,” *NeurIPS*, Dec. 2026.
- [C21] Jihun Yun, Sungjoon Yoon, Beomhan Baek, Minhak Song, **J. Jon Ryu**, Kwang-Sung Jun, “Beyond Clipping: Signed Logarithmic Smoothing for Policy Optimization,” *NeurIPS*, Dec. 2026.
- [P1] Tejas Jayashankar*, **J. Jon Ryu***, Xiangxiang Xu, Gregory W. Wornell, “Lifted Residual Score Estimation”. Submitted. (Note: A preliminary version of this manuscript is to be presented at *ICML 2024 Workshop on Structured Probabilistic Inference & Generative Modeling*.)
- [P2] **J. Jon Ryu**, Young-Han Kim, “An Information-Theoretic Proof of Kac–Bernstein Theorem,” arXiv:2202.06005.
- [P3] **J. Jon Ryu**, Young-Han Kim, “Minimax Optimal Algorithms with Fixed- k -Nearest Neighbors,” arXiv:2202.02464v3.

SELECTED
PREPRINTS

INVITED
TALKS

- **From Information Theory to Machine Learning Algorithms: Two Vignettes.**
 - Signals, Information and Algorithms Laboratory, MIT, Cambridge, MA, USA, Mar. 2022 (remote).
 - Center for AI and Natural Sciences, KIAS, Seoul, South Korea, Mar. 2022 (remote).

- **From Wyner's Common Information to Learning with Succinct Representation.**
 - Information Theory and Applications (ITA) Workshop, La Jolla, CA, USA, May 2022.
 - Machine Intelligence and Data Science Laboratory, Seoul National University, Seoul, South Korea, Jan. 2023.
 - Inference and Information for Data Science Lab, KAIST, Daejeon, South Korea, Jan. 2023.
- **Nearest Neighbor Density Functional Estimation From Inverse Laplace Transform.**
 - Center for AI and Natural Sciences, KIAS, Seoul, South Korea, Aug. 2022 (remote).
- **On Confidence Sequences from Universal Gambling.**
 - Prof. Aaditya Ramdas' Group Meeting, CMU, Pittsburgh, PA, USA, Oct. 2022 (remote).
 - Hanyang University, Seoul, South Korea, Jan. 2023.
- **Operator SVD with Neural Networks via Nested Low-Rank Approximation.**
 - MLTea talk, MIT, Cambridge, MA, USA, Nov. 2023.
 - Information Theory and Applications (ITA) Workshop, La Jolla, CA, USA, Feb. 2024.
 - Mitsubishi Electric Research Laboratories, Cambridge, MA, USA, Sep. 2024.
 - KAIST AI Graduate School of AI, Daejeon, South Korea, Sep. 2024 (remote).
 - Center for AI and Natural Sciences, KIAS, Seoul, South Korea, Oct. 2024 (remote).
 - Flatiron Institute, New York, NY, USA, Oct. 2024 (remote).
 - MIT JTL Urban Mobility Lab, Cambridge, MA, USA, Sep. 2025 (remote).
 - Prof. Devavrat Shah's Group Meeting, MIT, Cambridge, MA, USA, Nov. 2025.
- **Efficient Generative Modeling and Operator Learning from First Principles.**
 - Frontier Research, Prescient Design, Genentech, New York, NY, USA, May 2025 (remote).
 - Department of EE, POSTECH, Pohang, South Korea, Jun. 2025 (remote).
 - Department of CSE, POSTECH, Pohang, South Korea, Aug. 2025 (remote).
- **Tools for Scalable and Reliable Scientific Inference from First Principles.**
 - Prescient Design, Genentech, Sep. 2025.

HONORS AND AWARDS

Spotlight Recognitions (top ~3% acceptance rate)

ICML 2024, ICML 2025

Top Reviewer Awards

NeurIPS 2023, NeurIPS 2025, ICLR 2026, ICML 2026

Departmental Fellowship Department of ECE, UCSD

Sep. 2015 – Jun. 2016

Kwanjeong Scholarship for Graduate Study Kwanjeong Scholarship Foundation, South Korea

Sep. 2015 – Jun. 2020

Kwanjeong Scholarship for Undergraduate Study Kwanjeong Scholarship Foundation, South Korea

Mar. 2010 – Dec. 2013

University Students Contest of Mathematics Korean Mathematical Society

- Among non-math majors: Gold prize (2010), Honorable mention (2009)
- Among math majors: Bronze Prize (2013)

REVIEWER SERVICES

- **Journal:** IEEE Trans. Info. Theory, IEEE J. Selected Areas Info. Theory, J. Comput. Phys., Trans. Mach. Learn. Res.

- **Conference:** ISIT (2017, 2023, 2024, 2025), ITW 2022, AISTATS (2022, 2023, 2024), ICML (2022, 2023, 2024, 2025), NeurIPS (2022, 2023 (top 10% reviewer), 2024, 2025 (top 10% reviewer)), ICLR (2024, 2026 (top-200 reviewer)).

INTERSHIP EXPERIENCE

Research Intern

Jun. 2019 – Dec. 2019

AI Research Group, Qualcomm Technologies, Inc.

- Researched deep learning based sequential models for speech processing.

Research Intern

Jun. 2018 – Sep. 2018

Deep Learning Team, SoC R&D, Samsung Semiconductor Inc.

- Developed a new information-theoretic representation learning principle.

TEACHING EXPERIENCE

Instructor (Penn State)

- A-I 410 AI Systems and Tools

Fall 2026

Instructor (MIT)

- **6.7800 Inference and Information** Spring 2024, Spring 2025
 - Designed and taught new research-related topics as special sessions as a co-instructor.
 - **Topics:** minimax bit prediction, universal inference and concentration, universal learning approach for Rock–Paper–Scissor machine, variational perspective on generative modeling.

Teaching Assistant (UCSD)

- **ECE 250 Random Processes** Winter 2017
- **ECE 154C Communication Systems** Spring 2017
 - Designed hands-on programming assignments for the class based on Julia.
 - **Topics:** Basic source coding and channel coding algorithms.
- **ECE 225B Universal Probability and Applications in Data Science** Spring 2018
 - Designed hands-on programming assignments for the class based on Python.
 - **Topics:** Lempel–Ziv probability assignment, context-tree weighting, and universal portfolio.
- **ECE 269 Linear Algebra and Applications** Winter 2019

MENTORING EXPERIENCE

Tejas Jayashankar, Ph.D. Student, MIT (now at Meta) Sep. 2022 – Jul. 2025

- Topic: Developing new generative modeling techniques.
- Outcome: Co-developed a new training technique for one-step high-quality image generators [C12].

Abhin Shah, Ph.D. Student, MIT (now at Five Rings) Sep. 2022 – Aug. 2024

- Topic: (1) Learning with fairness. (2) Learning principles for unnormalized probability models.
- Outcome: Co-authored a paper on learning under fairness constraint with uncertain attributes, presented at ISIT'24. Co-authored a paper on unifying learning principles for unnormalized probability models [C13].

Maohao Shen, Ph.D. Student, MIT Sep. 2022 – present

- Topic: (1) Efficient uncertainty quantification algorithms. (2) Efficient LLM alignment with feedback data.
- Outcome: Co-authored a paper on uncertainty quantification for black-box models, presented at NeurIPS'24 [C11]. Developing a unified RLHF framework for LLM alignment (work in progress).

Axel Adjei, M.Eng. Student, MIT Sep. 2025 – May 2026

- Advised a master thesis on gradient-based iterative methods for spectral decomposition.
- Thesis: Gradient-Based Algorithms for Eigenvalue Decomposition: An Orbital Minimization Approach.

Pavan Yeddanapudi, Undergraduate Student, MIT Sep. 2024 – Apr. 2025

- Topic: (1) Efficient estimation of information measures for high-dimensional data [C17]. (2) Score-based techniques for representation learning (work in progress).

Samuel Zhou, Undergraduate Student, MIT Mar. 2025 – May 2025

- Topic: (1) New linear-algebraic optimization framework for representation learning [C16]. (2) New techniques for interpretable representation learning (work in progress).

OTHER EXPERIENCE

Military Service (mandatory) Mar. 2011 – Dec. 2012
Republic of Korea Army

REFERENCES

Gregory W. Wornell
Professor of EECS, MIT
Postdoc Advisor
✉ gww@mit.edu

Young-Han Kim
Professor of ECE, UCSD
Ph.D. Advisor
✉ yhk@ucsd.edu

Sanjoy Dasgupta
Professor of CSE, UCSD
Ph.D. Advisor
✉ sadasgupta@ucsd.edu